

positions, while (ii) *nondirectional* or *market-neutral* HFs try to extract value from “arbitrage” opportunities. Although the term is widely used in industry, these funds usually do not find pure arbitrages, which are rare. Rather, this type of HF seeks to neutralize market movements and attempts to profit from securities that are misvalued relative to each other.

Within these two approaches, there are two main styles of investing: (i) *discretionary* styles rely mostly on a trader’s judgment (or lack thereof), and (ii) *systematic* styles are more rules-based and rely on quantitative models. The quant funds that melted down in August 2007 fall into the latter category.

Lo (2007) suggests the quant fund losses reflect an underlying source of systematic risk. That most HFs did not deliver absolute returns over the financial crisis of 2008–2009 and instead posted large losses—whether they were market timers or nondirectional traders and whether they had discretionary or systematic styles—also suggests that all HFs, not just quant HFs, are exposed to common risk factors. In fact, *HFs are bundles of standard risk factors*.

The two most important factors are plain old. . . .

Equity Market and Volatility Risk

Table 17.5 reports partial correlations of the HF index and various HF strategies (from HFR) to the S&P 500 and a volatility factor. The latter is compiled by Merrill Lynch and is a return series from a short volatility strategy (see chapters 2 and 7). Short volatility strategies sell volatility insurance and make money during stable times but lose money when volatility spikes. I take monthly data from January 2000 to September 2012 and construct excess returns for the HF returns and the S&P 500 using U.S. T-bills as the risk-free rate. The partial correlation of the HF excess returns with the equity factor controls for the effect of the volatility factor and vice versa.

Table 17.5 shows very high correlations of all HF strategies with the market factor. The partial correlation of the HF index with respect to equity risk is a very high 0.67. HFs with emerging markets and event-driven strategies also have partial correlations above 0.6. Event-driven HFs take advantage of corporate transactions like mergers, restructurings, buybacks, security issuance, and so on. The distress, or restructuring, strategy invests in corporate bonds of companies that are in or close to bankruptcy. The partial correlation of distress with equities is 0.41. Only for the long-short equity funds, of which a large number are quant funds, is the partial correlation with equity market risk low at 0.11 and statistically insignificant. (It does seem that, on average, market-neutral HFs are market neutral.³²)

The partial correlations of HF returns with the volatility risk factor are somewhat smaller than the market, but Table 17.5 reports that they are still quite large.

³² Patton (2009) argues, however, that at the individual HF level, more than a quarter of long-short HFs exhibit statistically significant and economically large exposure to the market.